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Lesson 9: Neural Networks

Technologies for Artificial Intelligence

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Agenda

- A boundary whose **shape** is learned, not chosen in advance
- One neuron is one line, and exactly where that runs out
- Forward propagation, and backpropagation derived right to left
- Softmax, activations, and the vanishing gradient
- Initialisation, optimisers, regularisation: the failure modes

Exercise 8, before we start

- What counts is **what the metric could see**, not the score itself
- The gap to watch for: a silhouette score taken as proof that a clustering is right, when it only rewards round, well-separated groups
- Today the same obligation returns in a new form: a network will reach a high training accuracy very quickly, and that number on its own says almost nothing

What a network adds

- Boundaries so far had a shape **fixed in advance**
- A line; axis-parallel boxes; a curve from a chosen kernel
- A network learns the **shape of the boundary itself**
- Everything else today makes that possible, or stops it going wrong
- No architecture tour, no hardware, no network call

Three problems, and a ceiling of 0.97

- **Acceptance test**: two axes, one pass/fail verdict
- **Two-channel drift**: the exclusive-or (XOR) function
- **Handwritten digits**: 1,797 images, ten classes, real
- The rig is **wrong on 3% of units**, so every score today is read against a ceiling of **0.97**
- A line scores **0.55** on acceptance and **0.93** on digits

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The perceptron, and a unit you already have

- Rosenblatt, 1958: weights, a bias, “output 1 if the sum is positive”
- Swap the threshold for a sigmoid: lesson 4’s logistic unit, `LogisticRegression`
- What is new: the unit becomes a **component** of a larger network
- Undecided at 0.5 means the weighted sum is zero
- A line in two dimensions, a hyperplane in general

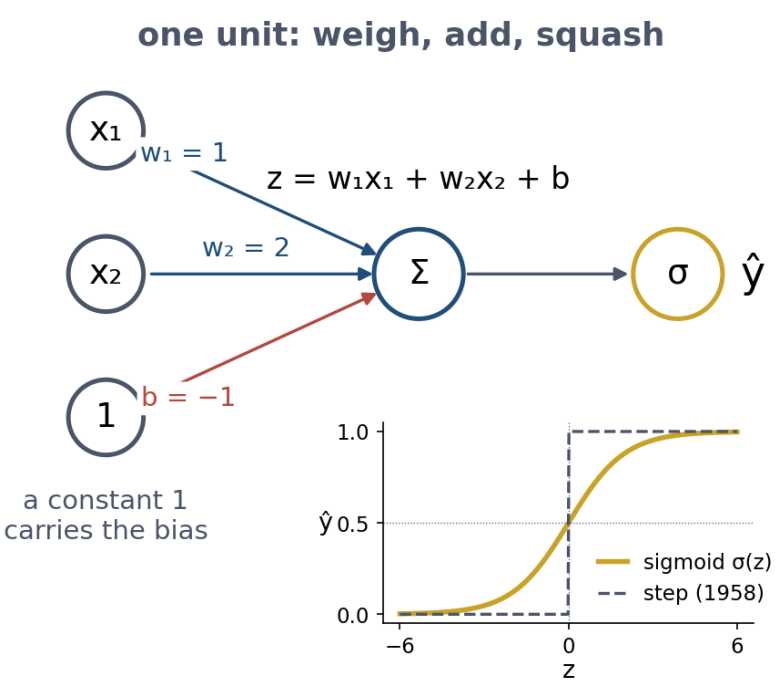
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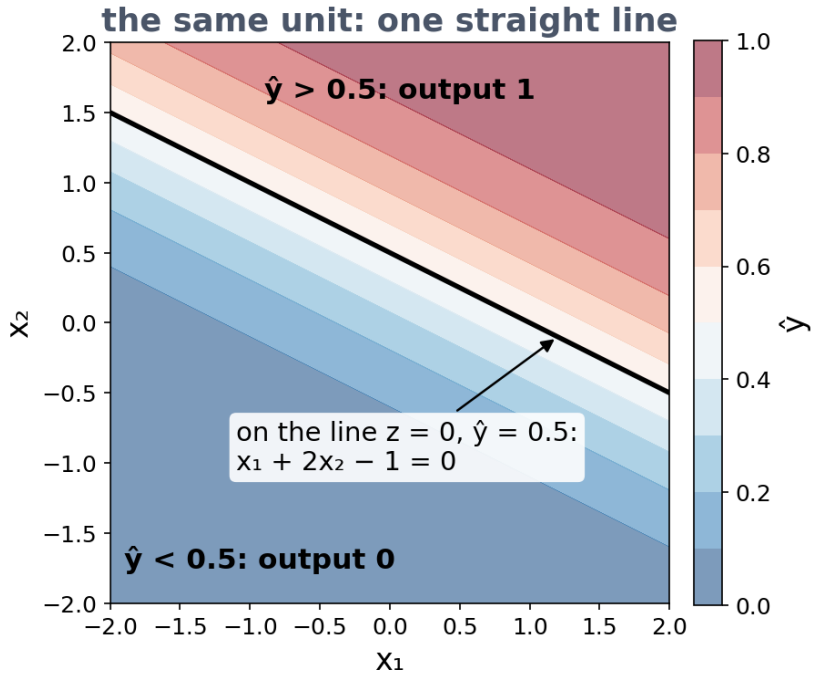
A neuron is a line

$$\hat{y} = \sigma(w^\top x + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}$$

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One neuron, drawn





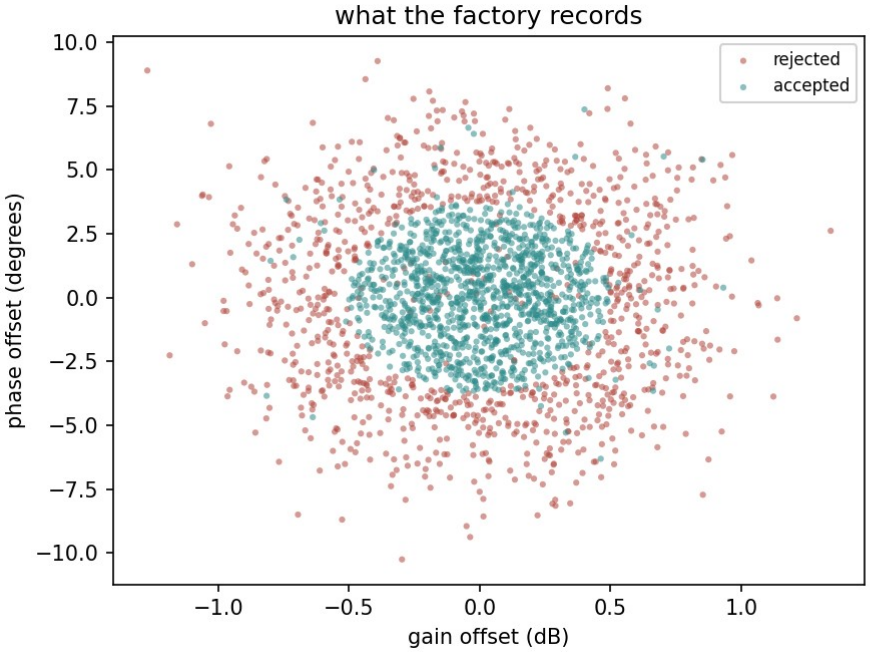
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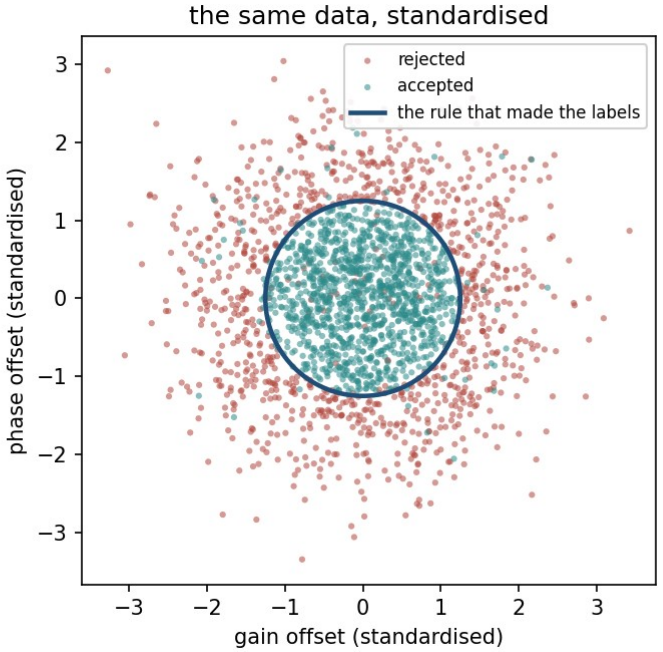
Meridian’s acceptance test

- Two calibration axes: a gain offset in decibels, a phase offset in degrees
- It passes when both offsets are **jointly** small enough for one global firmware correction
- 3,000 sensors: `train_test_split` keeps 2,250 to train, then `StandardScaler`
- So the accept region is the inside of a **closed curve**
- And no straight line encloses anything

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2,250 sensors, raw and standardised





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A circle of radius 1.25

- Tolerances **0.50** dB and **3.75**°; production spreads (standard deviations) **0.40** and **3.00**
- Both are exactly **1.25 spreads**, so standardised, the accept region is a circle of radius 1.25
- Production offsets are **normally distributed**: standardised, each unit is a random point of a standard two-dimensional normal
- **0.5497** of the 3,000 generated units fall inside the circle

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Before any model: the share inside, from the normal

$$P(\|z\| < s) = 1 - e^{-s^2/2}, \qquad s = 1.25 : \ 0.5422$$

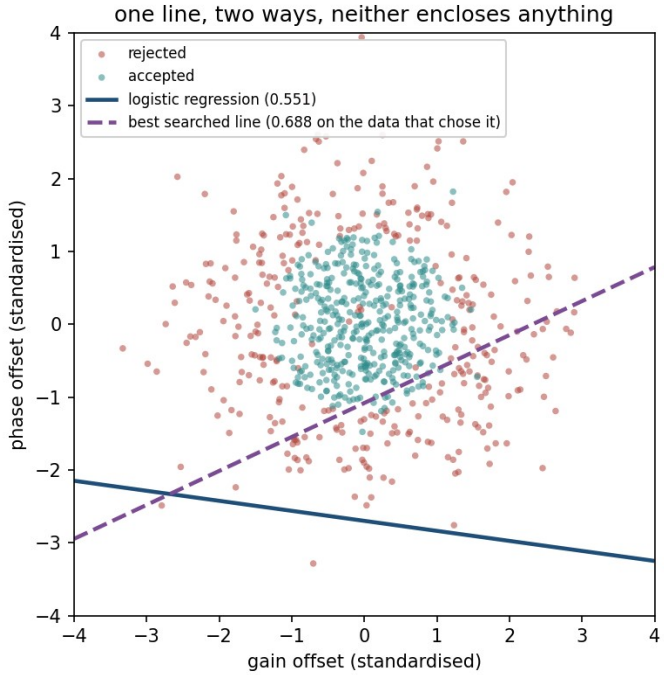
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What one line is worth

	accuracy
always predict “accepted”, on the 750 test units	0.5480
fitted logistic regression, on the 750 test units	**0.5507**
the best possible straight line, computed from the true distribution	0.6491
the rig’s ceiling	0.9700

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No line encloses it



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The fit did not fail

- Fitted logistic regression: coefficients **(0.0096, 0.0701)**, intercept **0.1892**, almost exactly the constant model
- Not an optimiser that gave up. The reason is symmetry
- The accept region is a disc on the origin; every accepted unit has a partner opposite
- Any tilt gaining accuracy on one side gives it back on the other

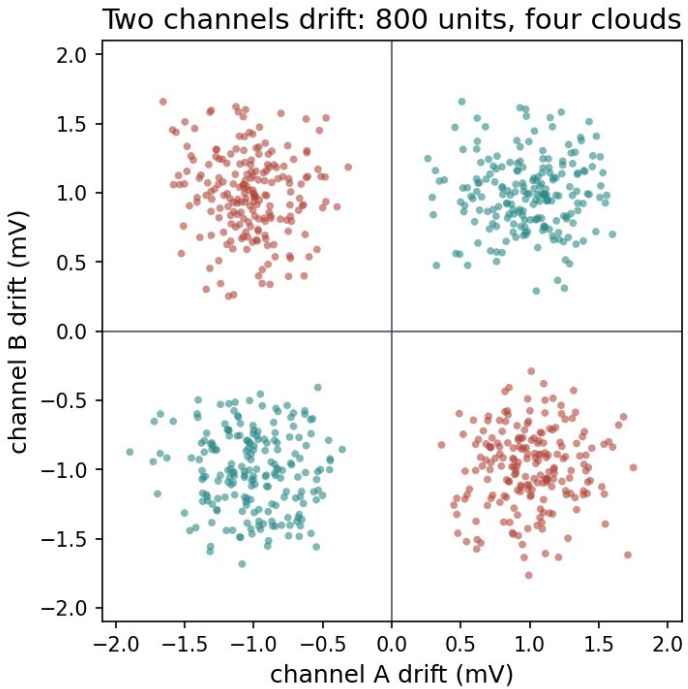
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72,762 candidate lines, and 4 points of self-deception

- 181 directions × 201 offsets × 2 sides = 72,762 lines, all scored on the test set; the winner, **on the data that chose it**: 0.6880
- The honest figure, computed exactly from the geometry: **0.6491**
- **Almost 4 points of pure selection bias**
- Nothing was fitted: only a choice made. Lesson 5’s argument again

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New data: two drifting channels



same sign (common mode): correctable
opposite signs (differential): not correctable

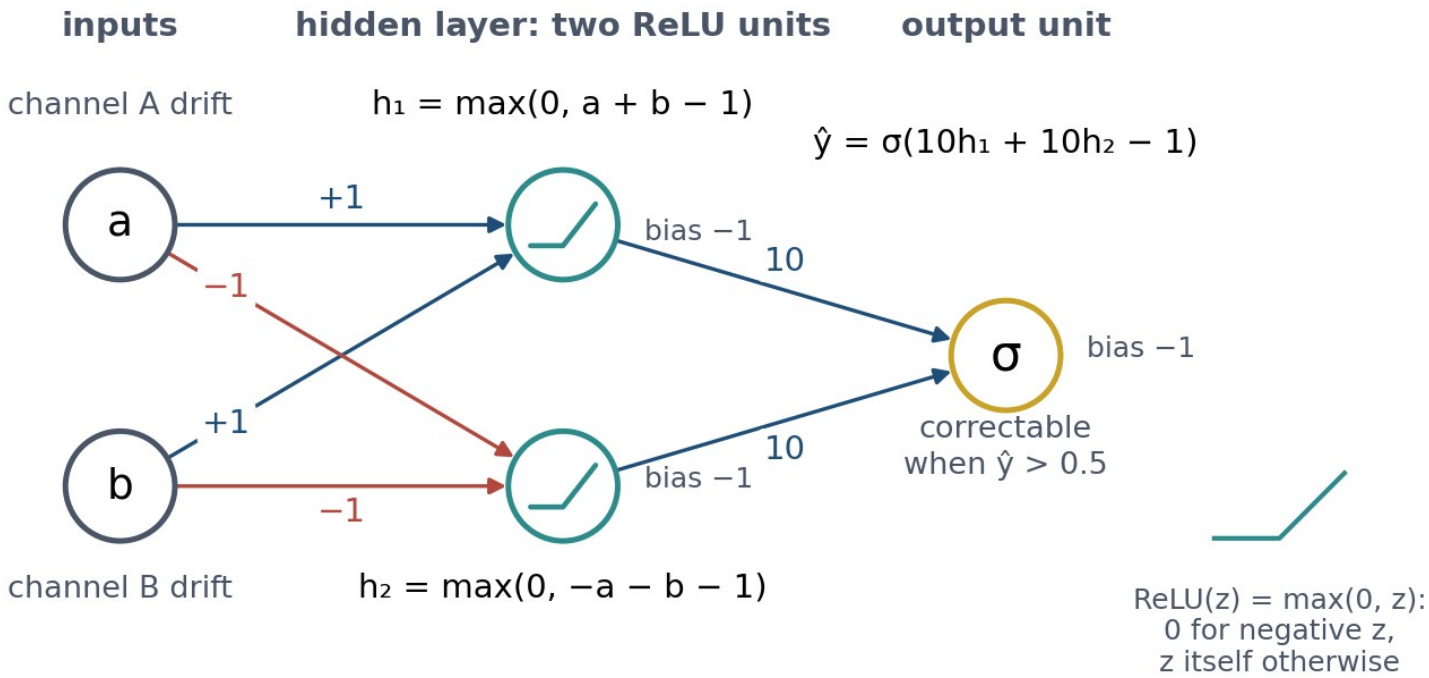
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The smallest problem that needs a hidden layer

- One correction removes the drift only when both channels drift **the same way** (same sign)
- That is the exclusive-or (XOR) function
- Four clouds of 200 units; a line scores exactly **0.5000**
- Coding the channels as ±1, the rule is about $|a + b|$
- “Large in absolute value” is the outside of a strip: **two boundaries, not one**

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Two hidden units, drawn

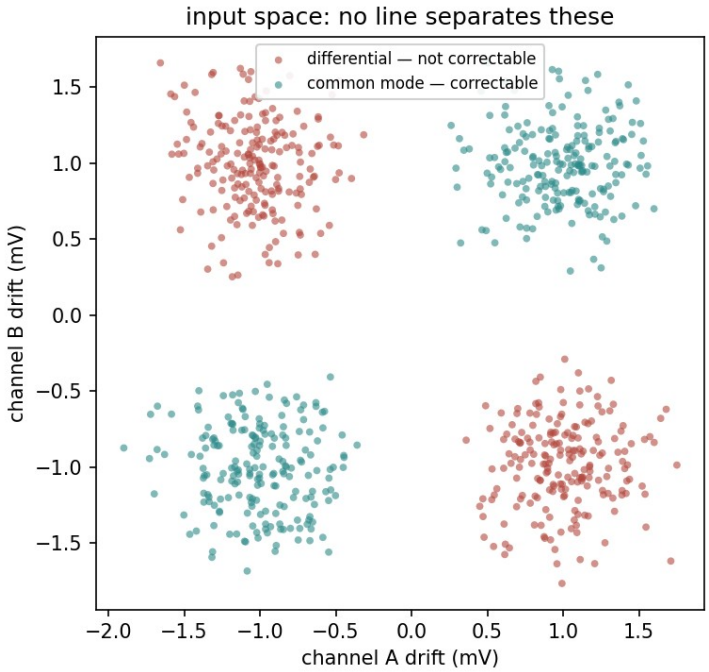


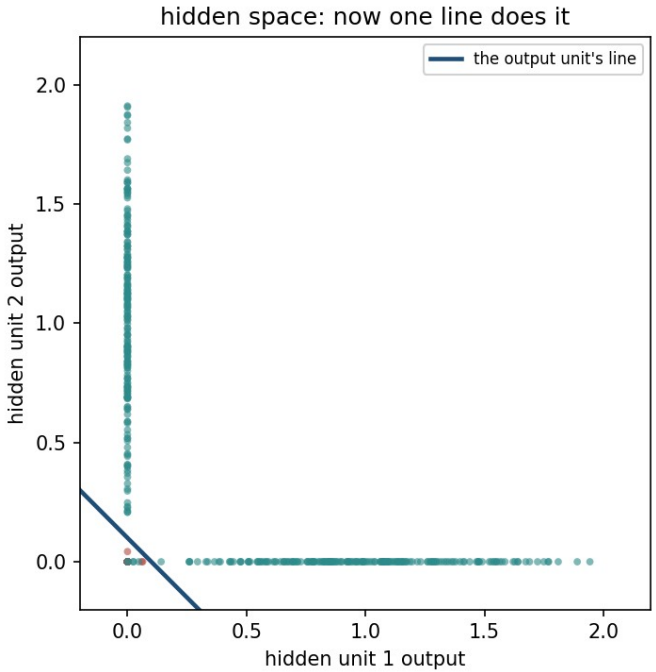
Two units, written down rather than trained

- Two rectified linear units, ReLU(z) = max(0, z), both fed the sum a + b
- One fires when a + b > 1, the other when a + b < -1
- Between them they compute |a + b|, clipped
- Output weights 10 and 10, bias -1: the rule is |a + b| > 1.1
- **Nothing trained**, and over all 800 units it scores **0.9938**

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What the hidden layer actually did



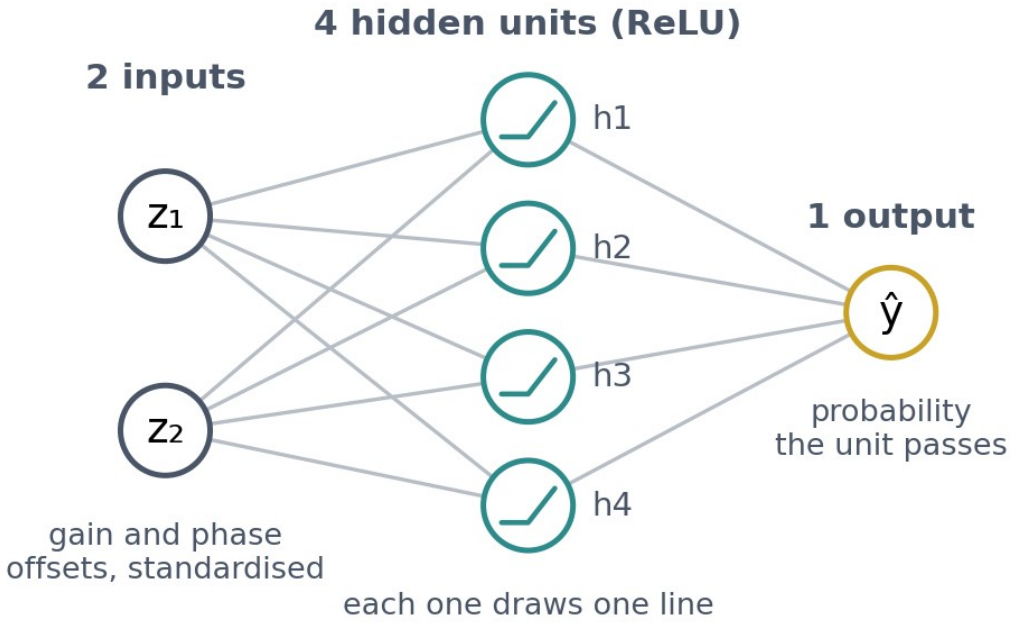


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The hidden layer classified nothing

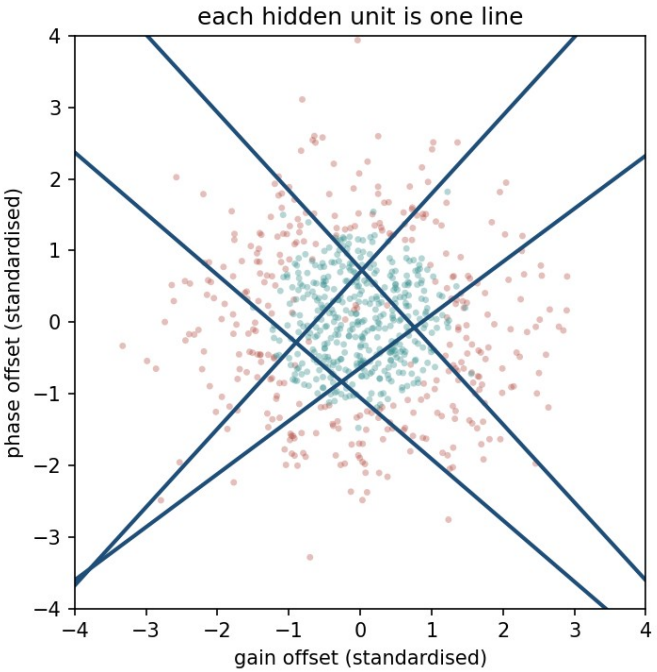
- It **moved the data** until the output line was enough
- Correctable clouds to (1,0) and (0,1); the rest to the origin
- The name for this is **representation learning**
- What is learned: coordinates in which the boundary is simple
- Lesson 10’s convolutional networks: the same trick, with most weights fixed at zero

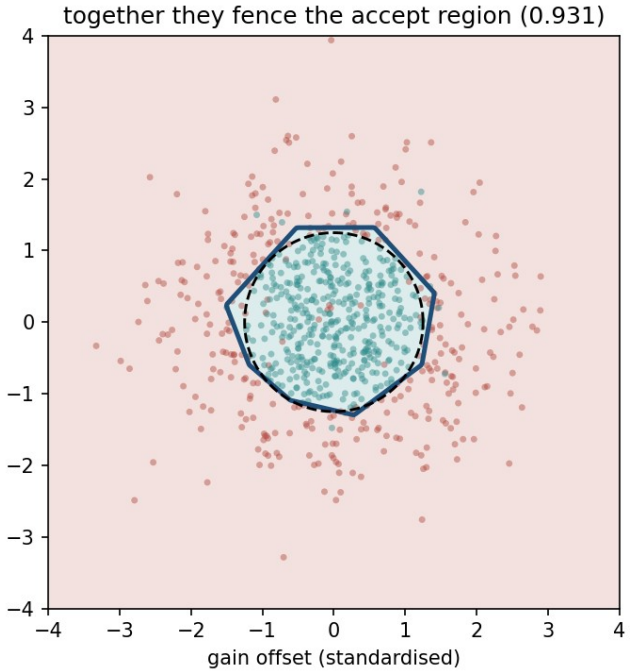
Back to acceptance: four hidden units



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Four lines enclose a region; one never does





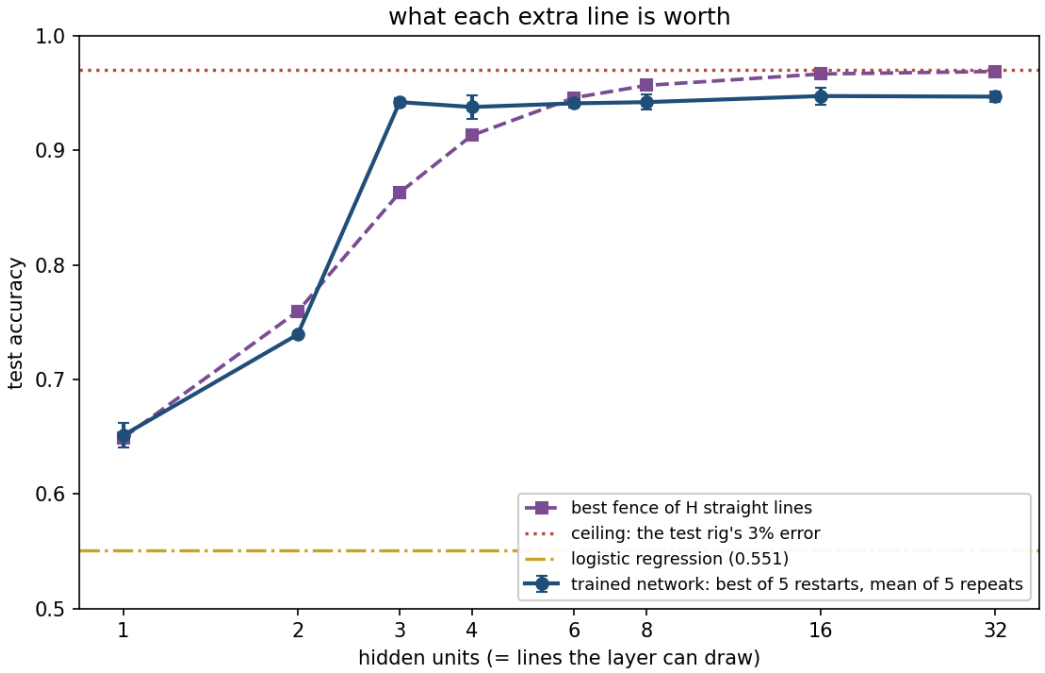
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Capacity is not the same as findability

hidden units (drift data)	median accuracy, 20 random starts	best of 20	starts above 0.95
2	0.7500	0.9975	**4 of 20**
3	1.0000	1.0000	15 of 20
4	1.0000	1.0000	19 of 20
8	1.0000	1.0000	20 of 20

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What each extra line is actually worth



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Universal approximation, and what it does not promise

- Cybenko (1989), Hornik (1991): one hidden layer, enough units, any continuous function
- Read the quantifiers. It says such a network **exists**
- Nothing about how many units “enough” is
- Nothing about whether training will find it
- Nothing about behaviour outside the region
- Two units sufficed on the drift data: descent missed it 16 times in 20

Forward propagation: examples in rows

- One hidden layer of H units, one output unit, m examples, n inputs
- Two matrix multiplies, each followed by an elementwise function
- Examples in **rows**, as scikit-learn and Keras; in columns every transpose flips
- In code: `keras.Sequential` of a `keras.Input` and `layers.Dense` layers

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Layer 1: a line per unit, then a non-linearity

$$Z^{[1]} = XW^{[1]} + b^{[1]}, \qquad A^{[1]} = g(Z^{[1]})$$

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Layer 2: one more line, then a probability

$$Z^{[2]} = A^{[1]} W^{[2]} + b^{[2]}, \qquad \hat{y} = \sigma(Z^{[2]})$$

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The shapes

symbol	shape	what it is
X	$m \times n$	the design matrix, one example per row
W, layer 1	$n \times H$	one column per hidden unit
b, layer 1	H	broadcast across rows
Z and A, layer 1	$m \times H$	pre-activations and activations
W, layer 2	$H \times 1$	the output unit’s weights
$\hat{y}$	$m \times 1$	one prediction per example

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Two facts the shapes hand you

- **Examples never mix**: the row index passes through untouched
- Nothing lets example 3 influence example 7, which is what makes mini-batching valid
- **Each column of the first weight matrix is one unit**
- Its weight vector, and the line that unit draws
- So H hidden units are H lines, and the output unit votes

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Untrained, the cost is 0.6721; predicting 0.5 for all, ln 2 = 0.6931

$$J = -\frac{1}{m}\sum_{i=1}^{m}\left[y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\right]$$

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Break

- Twelve minutes

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Two bad ways to get every gradient

- We need the derivative of the cost for **every** weight and bias
- **Perturb and re-run**: one forward pass per parameter; for notebook 3’s network of 301,066 parameters, 301,066 passes for one step
- **Differentiate symbolically**: the same sub-expressions, thousands of times
- Both correct. Both unusable

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Compute each repeated piece once, right to left

- Carry **δ**, the cost's derivative with respect to each layer's **pre-activations** Z
- From it: that layer's weight gradients **and** the layer below's, so one backward sweep gives every gradient
- **A weight's gradient is how wrong the layer above was, times what this weight contributed**
- Keras automates it with `tf.GradientTape`

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The output layer collapses to prediction minus truth

$$\delta^{[2]} = \frac{1}{m}(\hat{y} - y)$$

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Why sigmoid and cross-entropy belong together

- The loss derivative carries a factor 1 over $\hat{y}(1 - \hat{y})$
- The sigmoid's derivative **is** $\hat{y}(1 - \hat{y})$
- They cancel exactly, leaving $\boldsymbol{\hat{y} - y}$
- Squared error keeps that factor, and it is near zero exactly when the network is **confidently wrong**

One step down, through the activation

- Weight gradient = incoming activations, transposed, times δ: for layer 2, $(A^{(1)})^T \delta^{(2)}$
- To step down: multiply δ by the weight matrix above, then elementwise (⊙) by the activation's derivative g′
- For the ReLU, g′ is 1 where the pre-activation was positive, 0 elsewhere: a unit silent going forward passes nothing back

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Backward through one layer

$$\delta^{[1]} = \left(\delta^{[2]}(W^{[2]})^{\top}\right) \odot g'(Z^{[1]})$$

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What backpropagation costs

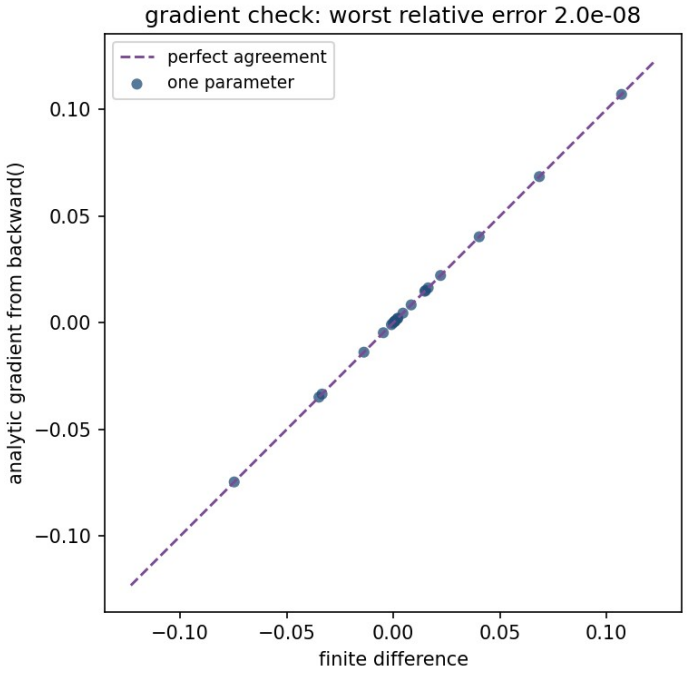
- The backward pass does about the same arithmetic as the forward pass
- **A gradient costs roughly what a prediction costs**: this is what makes training feasible at all
- But every layer’s activations must be **stored**
- Memory grows with depth times batch size, and memory, not arithmetic, is what usually limits batch size

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Check the gradient before you trust it

- A wrong gradient **raises no exception**: it just trains badly
- Compare each partial with $(J(\theta + h) - J(\theta - h)) \div 2h$: error of order h^2
- Relative error = $|\text{numerical} - \text{analytic}| \div (|\text{numerical}| + |\text{analytic}|)$; notebook 01: worst 1.97×10^{-8}, median 7.83×10^{-11}
- **Below 10^{-6}: believe it. Above 10^{-4}: there is a bug**

21 partial derivatives, on the diagonal



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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
split, then one straight line, 9-16	`train_test_split`, `StandardScaler`, `LogisticRegression` → 0.5507
forward, backward, gradient check, 28-43	NumPy only → untrained cost 0.6721, worst disagreement 1.97×10^{-8}
the drift problem, 17-25	the hand-built network, then 20 restarts per width → 4 of 20 above 0.95 at two units
the polygon yardstick, 26	`scipy.integrate` → the best 4-sided fence scores 0.9396
zero initialisation, 56-57	all weights 0 → final cost 0.6887

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Ten classes need ten outputs

- **Softmax** turns K scores into a probability distribution
- For K = 2 it is the sigmoid: one construction, two sizes
- Loss, categorical cross-entropy: −log(probability of the true class), `sparse_categorical_crossentropy`
- The gradient at the output is again $\hat{\mathbf{y}} - \mathbf{y}$: the backward pass is unchanged

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Softmax

$$\text{softmax}(z)_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}$$

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A hidden layer is not always the answer

architecture, on the digits	parameters	validation accuracy	sd over 3 seeds
softmax alone, no hidden layer	650	0.9324	0.0035
one hidden layer of 32	2,410	0.9602	0.0052
one hidden layer of 64	4,810	0.9611	0.0045
two hidden layers of 64	8,970	0.9676	0.0057

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Reading that table

- No hidden layer at all is within **3.5 points** of the best network here
- The first hidden layer is worth **2.8 points**; 32 units to 64, **0.09**
- On the acceptance data the same step was worth **39 points**: 0.55 for the line, about 0.94 with a few hidden units
- Digits vote pixel by pixel: the acceptance rule needed two measurements **at once**

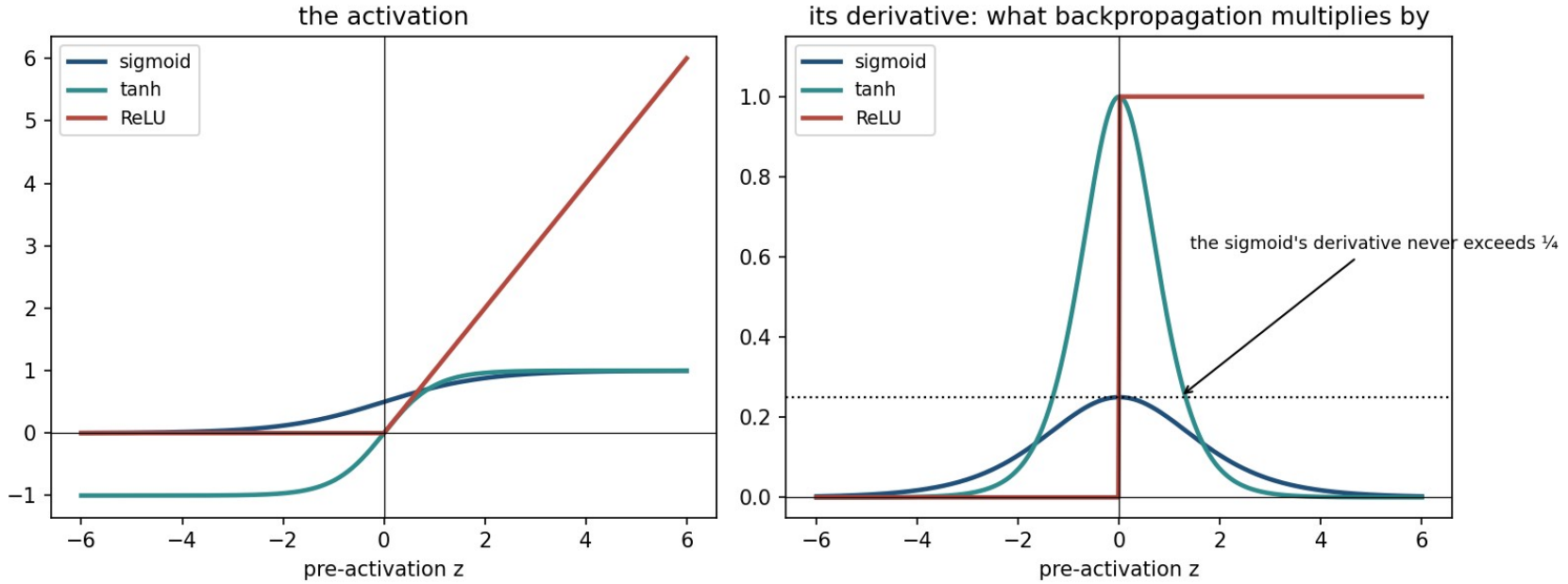
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A sigmoid layer divides the gradient by about four

- Any number of linear layers composes to one: the activation is the whole reason depth buys anything
- The sigmoid’s derivative $\sigma(z)(1 - \sigma(z))$ peaks at **¼**, at zero only
- Backpropagation multiplies by it **once per layer**
- At initialisation the weights multiply it by about 1, so the derivative decides: **each sigmoid layer divides the gradient by about four**

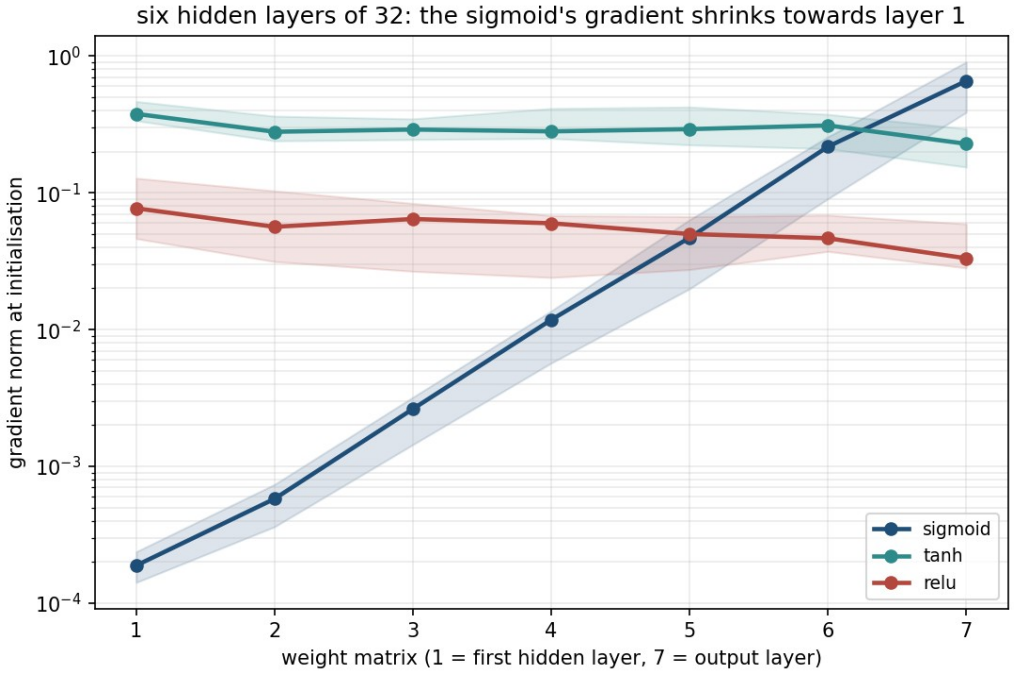
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Three activations, and their derivatives



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Six layers of it, measured



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One draw is not a law

- Six sigmoid layers, eight seeds: gradient size at the layer nearest the output ÷ at the first layer, median **3,547**
- Over those eight seeds it ranges **2,734 to 4,607**
- Quoting the largest reports one draw as a law
- **The per-layer factor of about 4 is the property**
- By depth, five seeds each: 9.5, 170, 3,553, 46,250 at two, four, six and eight layers

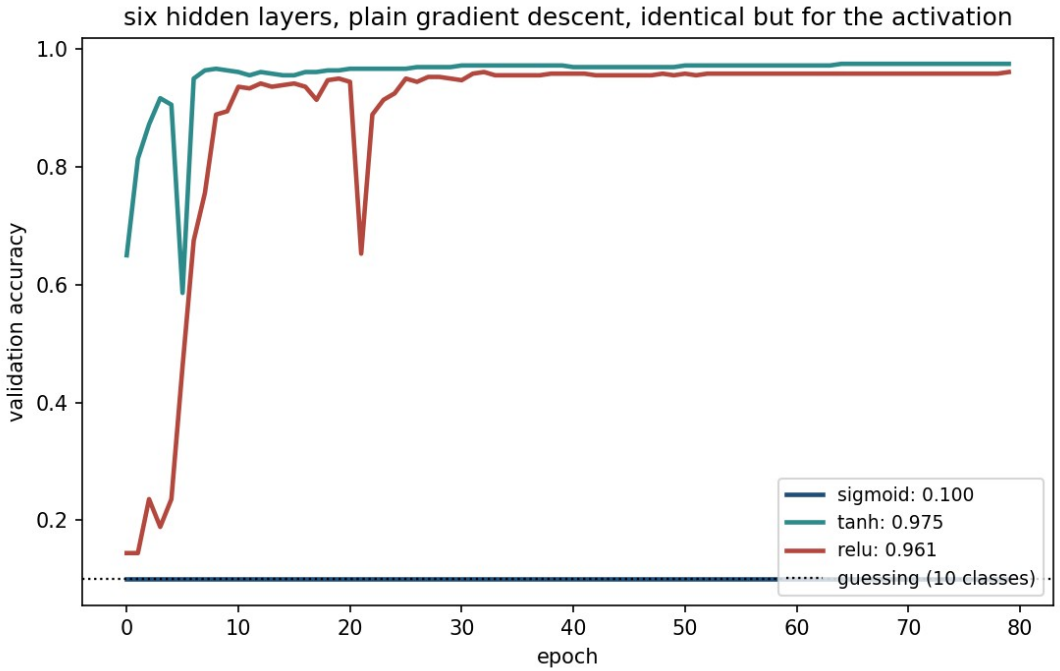
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Eighty epochs, three activations: only the sigmoid never learns

- Same six layers, same data, differing **only** in the activation
- Sigmoid: **0.1000**, exact chance on ten classes
- tanh: **0.9750**, past 0.85 within 3 epochs
- Rectified linear unit: **0.9611**, 9 epochs to get going
- tanh squashes too, but its derivative reaches 1; the sigmoid’s never exceeds ¼, and that is what kills it

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The sigmoid never leaves chance



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Notebook 2, live

section, and its slides	you call (→ the number to come back with)
the same network in Keras, 28-33	`keras.Sequential`, `keras.Input`, `layers.Dense`, `SGD`; then `compile`, `fit`, `evaluate` → 0.9333
ten classes, 45-48	`Dense(10, activation="softmax")`, `Adam`, `Dropout`, `confusion_matrix` → 0.9676 with two layers of 64
the shrinkage, measured, 49-52	`tf.GradientTape`, `sparse_categorical_crossentropy` → about 4 per sigmoid layer
training six layers, 53-54	`SGD` on three activations → sigmoid 0.1000, tanh 0.9750
how big the weights should be, 58	`RandomNormal` at two scales, against Glorot → 73% saturated at N(0, 1)

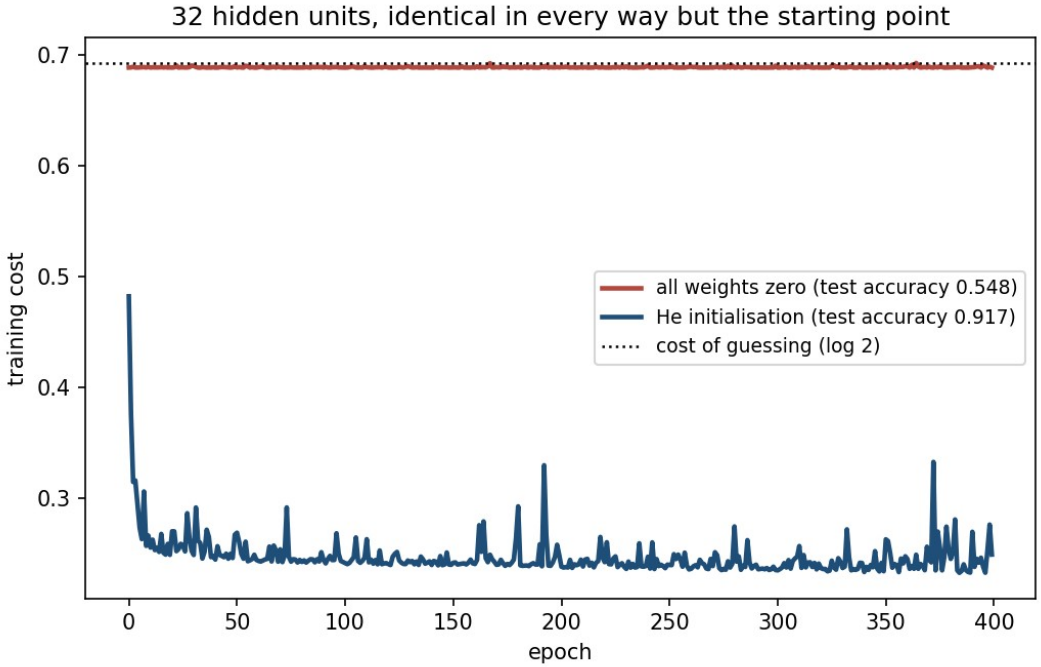
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Zero cannot work

- Lesson 3 started from zero, and was right to: those costs are convex
- All-zero weights **freeze** a network: **only the output bias moves**
- It converges to the base rate, the cost to the label entropy: **0.6887 predicted, 0.6887 measured**
- 32 hidden units started at zero end as **1 distinct unit**, copied 32 times

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Only the starting point differs



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How large? Propagate the variance

- Each layer multiplies the variance by $n \times \mathrm{Var}(w)$: $\mathrm{Var}(w) = 1/n$ is **Glorot**; for ReLU, $2/n$ is **He**
- Eight tanh layers of 64, weights of standard deviation 0.01: the signal **collapses** to zero by layer 4
- Standard deviation 1: 73% of the last layer **saturates**, $|a| > 0.99$
- Glorot keeps half its spread over the eight layers

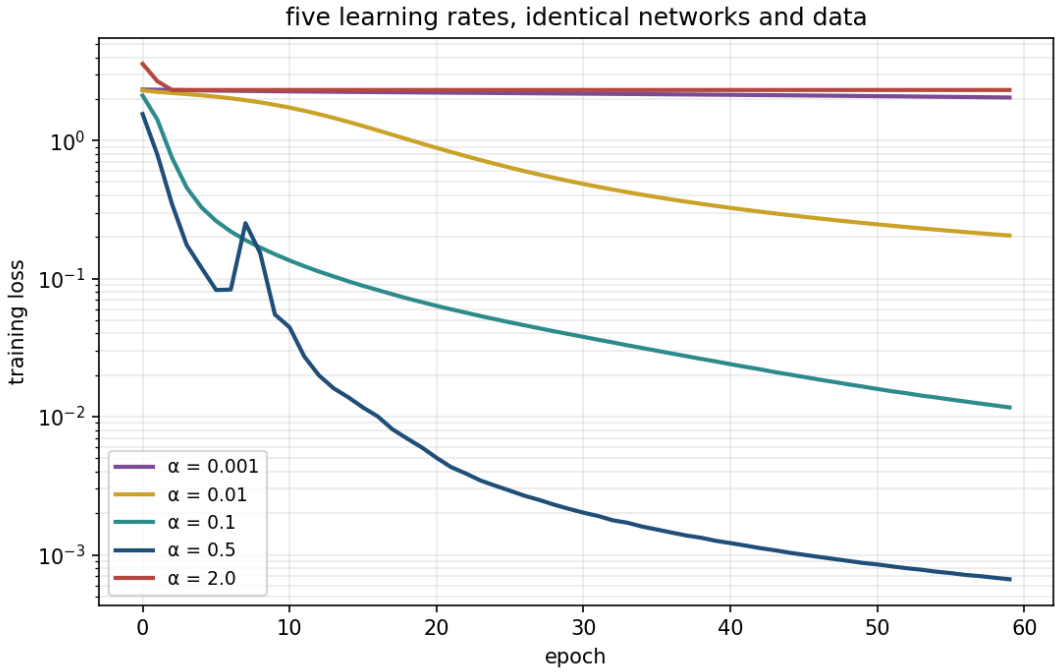
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The learning rate, swept: two layers of 64, on the digits

α, plain SGD, 60 epochs	final training loss	validation accuracy	sd over 3 seeds
0.001	1.9550	0.5574	0.0794
0.01	0.1965	0.9398	0.0035
0.1	0.0115	0.9556	0.0039
0.5	0.0007	**0.9685**	0.0026
2.0	2.3222	0.1009	0.0013

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Three regimes, and a narrow band between them



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The predictable mistake

- Too small, and the loss is still falling when the epochs run out
- **It looks exactly like a network that is too small**: both accuracies low
- The instinct that follows is sound: low training accuracy really is the signature of underfitting
- **Vary α over orders of magnitude before touching the architecture**

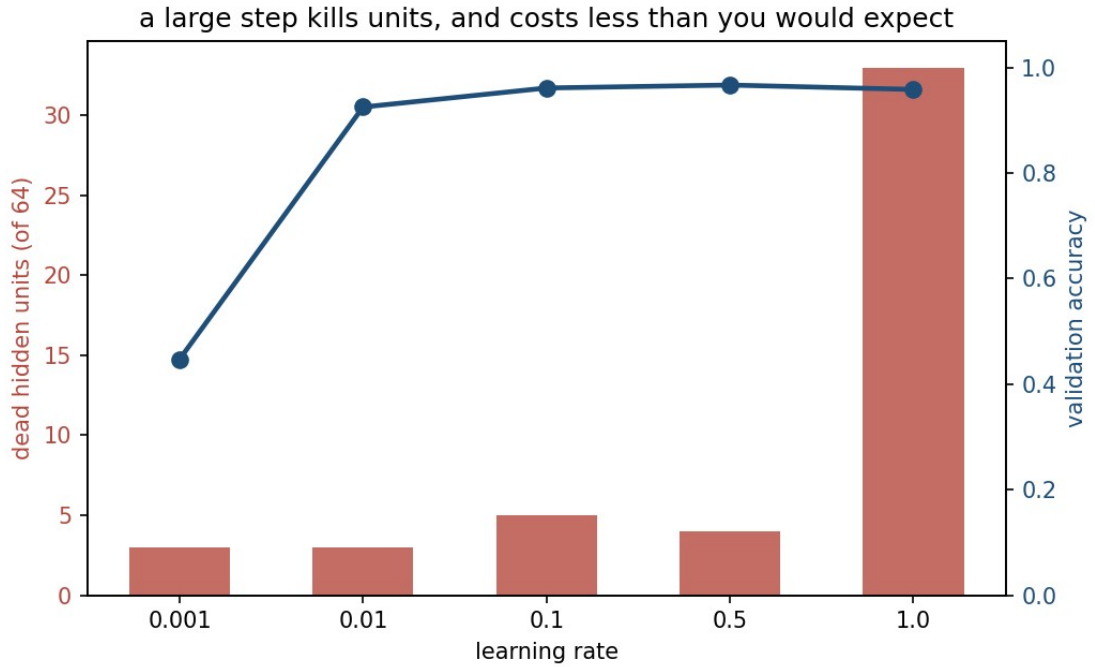
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The rectified linear unit has its own failure

- Its derivative is exactly 1 on the active side: so, no vanishing gradient
- But a unit negative for **every** example gets zero gradient, for ever
- **Dead**: the gradient that would revive it is the one it cannot receive
- One layer of 64, 40 epochs: at $\alpha = 1.0$, **33 of 64 are dead**, validation accuracy 0.9583 against 0.9667 at $\alpha = 0.5$

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Half the layer dead, and almost no visible cost



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Mini-batches, momentum, and Adam

- Mini-batch **stochastic gradient descent (SGD)**, `SGD`: a small batch's gradient, noisy but right on average
- **Momentum**: step along a running average of past gradients
- **Adam (adaptive moment estimation)**, `Adam`: each parameter's step scaled by its own recent gradients, so a badly chosen α matters less

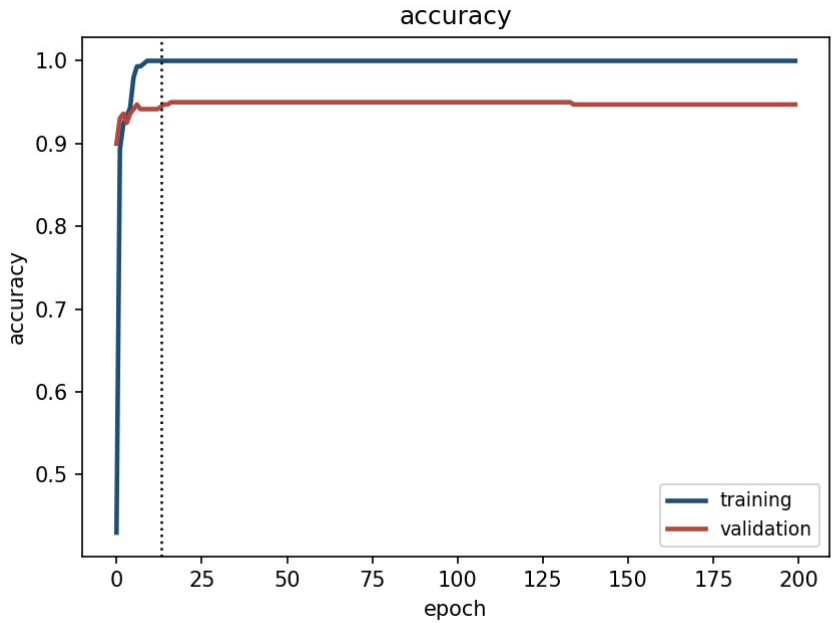
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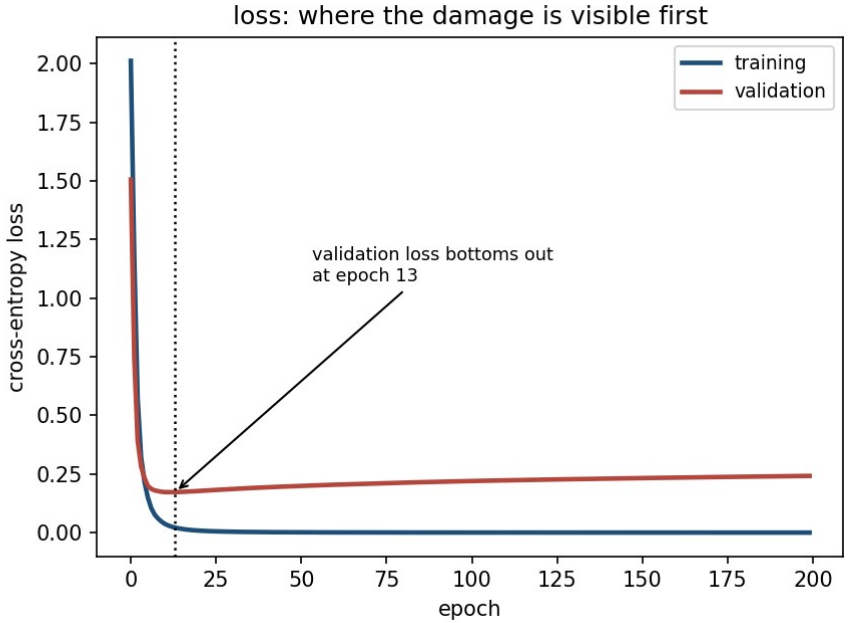
Adam, measured: read the spread before the mean

optimiser, acceptance data, 16 hidden units	test accuracy	sd over 5 seeds	worst of 5	against the noise-free labels
plain gradient descent	0.9389	0.0184	0.9027	0.9685
with momentum 0.9	0.9349	0.0094	0.9173	0.9645
Adam	**0.9485**	0.0072	0.9360	0.9792

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It memorises the training set, and generalises anyway





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Three regularisers on that network

method	test accuracy	sd over 5 seeds	train − test gap	epochs run
early stopping only (`EarlyStopping`: keep the best epoch)	0.9411	0.0069	0.0589	38.8
+ dropout 0.4 (`Dropout`: drop 40% of units per batch)	0.9483	0.0038	0.0503	48.8
+ L2 10^{-3} (`l2`: a penalty on squared weights, as Ridge)	0.9478	0.0011	0.0522	198.4
+ dropout and L2	**0.9511**	0.0045	0.0489	161.6

Notebook 3, live

section, and its slides	you call (→ the number to come back with)
the learning rate, 59-61	`SGD(learning_rate=α)` at five rates → best 0.9685 at α = 0.5
dead units, 62-63	`keras.Model` to read the hidden layer → 33 of 64 dead at α = 1.0
optimisers, 64-65	`SGD`, `SGD(momentum=0.9)`, `Adam` → Adam's worst run 0.9360
memorising, 66	two layers of 512 on 300 digits, `count_params` → 301,066 parameters
regularisers, 67	`EarlyStopping`, `Dropout(0.4)`, `l2(1e-3)` → 0.9411 to 0.9511
more data	`train_test_split` at six sizes → +3.2 points from 300 to 1,077 examples

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What to take away

- **A neuron is a line**: three enclose a region, one never does
- A hidden layer **moves the data**; it does not classify
- **Backpropagation is the chain rule** - so gradient-check it
- **A sigmoid layer divides the gradient by about four**
- Report a difference with its spread

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Homework

- **Exercise 9**, discussed at the start of **Friday 4 December**
- `Exercises/09_neural_networks.md`

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